ICT302 IT Professional Practice Project

Project Name:

Secure QR Code Development

Client Organization:

Secure QR Code Project Client

Group Name:

VAFPQR

PROJECT MANAGEMENT PLAN

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Table of Contents

1. Executive Summary **5**
2. Introduction **6**
3. Project Integration Management **7**
4. Project Scope **10**
4.1.1 Project Goals 10
4.1.2 Project Benefits **11**
4.1.3 Project Deliverables **12**
4.1.4 Scope Statement 13
4.2 Work Breakdown Structure Summary **16**
5. Project Time Management **19**
5.1 Project Milestone Dates 20
5.2 Timeline for Development Work 20
5.3 Gantt Chart 21
5.6 Buffer Days Prior to Deadlines 22
5.7 Recording Actual Finish Dates Against Planned Finish Dates 23
6. Project Quality Management **24**
6.1 Component quality checklist 24
6.2 Quality assurance testing 24
6.3 Performance metrics, integrity safeguards and acceptance criteria 26
7. Project Communications Management **30**
7.1 Stakeholder Analysis 30
7.2 Communication Methods and Frequency 32
7.3 Response Time and Communication Rules 34
7.4 Tone of Communication 34
7.5 Documentation Formatting 35
7.6 Technologies and Access Methods 36
8. Project Human Resources Management **38**
9. Project Risk Management **45**
9.2 Probability/impact matrix 57
10. Project Cost Management **58**
10.1 Cost Estimation Method 58
10.4 Cost Control 62
12. Conclusion **67**
13. Appendices **68**

13.1 Appendix A: Deliverable Task Breakdown Statement.....68
13.2 Appendix B: Work Breakdown Structure and Schedule.....70
13.3 Appendix C – Project/Team charter.....78
13.4 Appendix D – Glossary of terms.....82
13.5 Appendix E – Agenda and minutes of all client meetings.....83
13.6 Appendix F – Agenda and minutes of all supervisor meetings.....86

Version Control

Version	Date	Description of Changes
0.1	25 May 2026	Initial Document Creation
1.0	6 June 2026	First submission
1.1	24 June 2026	Feedback received and revised document accordingly
1.2	20-31 July	Final document revision and changes were made (ES, methodology, scope, milestone, quality management, cost management, glossary, formatting and diagrams)
2.0	1 Aug 2026	Final Submission

1. Executive Summary

The VAFPQR Project Team's Project Management Plan (PMP) is presented in this document. It outlines how our six man team, Patricia Orenza (Project Manager), Felixia Lim, Roy Hojin, Gu QiYu, Alvin Yeow, and Vanessa Tan will plan, carry out, and deliver the Secure QR Code Authentication System for clients Peter Cole and Mike Groeneweg between May 20 and August 1, 2026.

An integrated management approach including scope, time, quality, communications, human resources, risk, cost, and procurement is established in this project management plan.

The Requirements and Analysis document, Design document, and final submission are among the main deliverables tracked against predefined milestone dates and supported by a Gantt chart and network diagram to manage task dependencies. Development is organized into sprints.

A stakeholder communication plan outlines procedures, frequency, and response-time expectations for the client and supervisor, while defined component checklists and acceptance standards ensure quality. Project risks are recognized and prioritized in a risk register backed by a probability/impact matrix, and roles and responsibilities are distributed throughout the team in accordance with the human resources plan. The labor and overall budget are estimated in a cost management area, while procurement needs are handled independently. When combined, these components provide a structured framework for the timely, cohesive scope delivery of a QR authentication platform ready for production.

2. Introduction

QR codes are now widely used in public signage, restaurant menus, payment terminals, and product packaging. However, its ease of use has created a serious, often overlooked security flaw. Fraudulent QR codes can be easily placed over or if not generated securely, can lead users to phishing pages, phony payment portals or harmful websites and these often come with no obvious warning that something is wrong. As QR code usage continues to grow across industries, so does the potential for exploitation.

This project directly addresses that gap. Businesses may create cryptographically secured QR codes, track their usage in real time, and receive automated notifications when tampering is discovered using the Secure QR Code Authentication System, a multi-layered platform. The system consists of three integrated parts, a web-based management dashboard that provides businesses with complete visibility into scan activity and security events, a mobile application that allows end users to scan and authenticate QR codes, and a backend API that handles code generation, validation, and data management.

With an emphasis on providing a production-ready system that is safe, scalable, and flexible across a variety of industries and use cases, the platform was created in close collaboration with clients Peter Cole and Mike Groeneweg. The design, architecture, development process, and results of the system are described in the document that follows.

3. Project Integration Management

Project Management Approach

Project initiation took place on 20th May 2026 through a kick-off meeting and the team's communication forum was established on the same day. The Team Charter was submitted by the team on 23rd May 2026. During this phase, the tasks were assigned to the team members, one drive was created by the team and communication processes were established.

The project management approach adopted by the team was a hybrid project management methodology, involving the linear planning approach of Waterfall as well as iterative approach of Agile. This approach was chosen because it includes the structured phases of Waterfall and the flexibility of Agile.

- Waterfall (Predictive) Components: A predictive, Waterfall-style approach was used for the high-level planning, system architecture definition, requirement finalization, and thorough project documentation, including the Project Management Plan, Requirements and Analysis Document, and Design Document. This created a precise baseline for the project's deliverables and scope, guaranteeing that important benchmarks were established and authorized prior to the start of development.
- Agile (Adaptive) Components: The development and implementation of the secure QR code system was carried out using Agile approach, which included sprints of two weeks each. The requirements for the project were collected directly from the clients, Peter Cole and Mike Groeneweg, analyzed by the team, and developed through several iterations of development and testing cycles.

The members participate in weekly online meetings that are generally held in the evenings and last for about two to three hours. In the process, each individual gives three major updates which include achievements after the previous meeting, activities planned for the next week, and the issues he/she is encountering. The minutes of the meeting are recorded in the OneDrive account.

The Excel spreadsheet is utilized to organize and document all the activities of the project. These activities are classified into four categories: To Do, In Progress, Completed, and Blocked. All activities that were extracted from the Work Breakdown Structure are documented including the deadline dates and the names of the members of the team who have been tasked with performing them.

Updates to the client are made once each major milestone has been accomplished. Some of the major milestones that have been identified include the approval of requirements, design, demo, system ready for testing, and product delivery. For every

milestone accomplished, an email update will be sent to the client, together with an update via Microsoft teams channels. The client is also granted access to the development environment, where they can view the system directly through video demos.

Upon completion of the project, the customer will receive all of the source code and documentation in PDF form, together with a training session lasting one hour using Microsoft Teams. The training will be about QR code creation and system account management.

Project Management Approach

Change is always expected in any project, and hence a three-tiered change control approach has been adopted.

1. Minor Changes (No significant impact): Changes that require two hours or less of work without having any effect on the scope of the project or the functionalities of the system can be done after notifying the team members. The person making the change needs to notify the other people through WhatsApp within 24 hours and update the change log.
2. Moderate Changes (2-8 hours of work): Any changes that require 2 to 8 hours of work involve a voting process for the team during the weekly meeting. A team member proposes the change, explains the effects of such a change, and gets the consensus of the whole team. Afterward, the task list and due dates for each change are recorded in the Excel project file.
3. Major Changes (Over 8 hours or affecting functionality): Any change that will take more than eight hours or involve functionality changes must have explicit approval from the client. This is done by sending an email to Peter and Mike stating why the change is necessary, how many extra hours or days it will take, and any existing functionality that needs to be dropped or put off. The client gives the approval through Microsoft Teams or email.

These processes cover risks R01 (Scope creep) and R11 (Schedule delay), as indicated on the project risk register.

The following items cannot be changed without prior permission from the client:

- JWT signing procedure for QR codes
- Blacklist for compromised codes
- Tampering alert procedure
- Database design once coding has commenced
- Final submission deadline

Project Management Methodology

In the process, a combination of both Waterfall and Agile methodologies is used.

1. Waterfall (Predictive) Planning: The first phase of planning, requirement analysis, and system architecture planning was done through the Waterfall or Predictive approach. This was to ensure that a good foundation is set up by good planning and documentation (R&A, PMP, Design).
2. Agile (Adaptive) Development: The development phase has been performed on the basis of an adaptive method with two-week sprints. The adaptive approach has been adopted due to its efficiency and applicability to a six-member team. The overall time frame of the project is from 20 May 2026 to 1 August 2026.

The two important tasks that occur at the end of each sprint cycle include:

1. Lecturer Feedback Meeting: The team shows off the application’s newly introduced features to the client through Microsoft Teams and obtains feedback on the performance of the sprint.
2. Sprint Retrospective: At the end of the sprint cycle, the team conducts a retrospective meeting where they ask themselves the following important questions:
    - What went well in the sprint?
    - What can be improved in the next sprint?

4. Project Scope
4.1.1 Project Goals

Academic Goals

- Apply real-world software development practices in a team environment, including requirements gathering, system architecture design, iterative development, and testing.
- Demonstrate competency in full-stack development, mobile development, and secure system design, including authenticated session management and role-based access control.
- Produce professional-grade documentation and deliverables that reflect industry standards, including a formal register of the differences between the baselined design and the delivered system.

Business Goals

- Deliver a secure QR code authentication system that protects businesses and the public from physical QR code tampering and fraud.
- Provide clients Peter Cole and Mike Groeneweg with a deployable academic prototype that is maintainable and designed to progress towards production readiness.

4.1.2 Project Benefits

Team Benefits

Gain hands-on experience building a multi-layered system spanning backend API development, mobile application development, and web-based dashboard design.

Gain practical experience implementing a complete custom authentication and authorisation layer (password hashing, single-use reset tokens, session revocation, and role-based route protection).

Develop professional collaboration skills including GitHub version control, task management, and client communication.

Build a portfolio-ready project demonstrating real-world technical competency.

Client Benefits

A complete platform for generating, managing, and authenticating QR codes with cryptographic security, restricted so that only authorised administrators can issue codes under the organisation’s trust umbrella.

User-assisted reporting of suspected physical QR code tampering, reducing fraud risk for businesses and their customers. Tamper reports are attributable to a signed-in account rather than anonymous.

A management dashboard providing server-backed visibility into QR code usage, scan metrics, tamper alerts, a curated recent-activity feed, and CSV export of the QR code register.

4.1.3 Project Deliverables

Documentation

Requirements and Analysis (R&A) Document

System Architecture Design Document

Project Management Plan including Scope, Schedule, and Risk Management

Design Change Register recording every material difference between the baseline documents and the delivered system

Test Plan and Test Results Report

User Manual

Installation and Deployment Guide

Final Project Report

Project Components

Backend API server (Node.js + Express + PostgreSQL via Prisma ORM), which also serves both web builds from a single hosted service

Two web-based surfaces (React):

QR Code Landing Page — displayed when a user scans a QR code with a regular scanner (Apple/Google). Validates the token server-side, displays the verification result, and prompts the user to download the dedicated app for full security features.

Admin Management Dashboard — a full single-page web application for administrators to manage QR codes, review tamper alerts, administer user accounts, and monitor scan metrics. The QR Code Display function is delivered inside this application as the generation-result modal and the QR code detail view, rather than as a separate public page.

Mobile application for QR code scanning, verification, personal scan history, and tamper alert reporting (React Native, Android-first, distributed as a direct APK download)

PostgreSQL database with a full five-table schema covering users, QR codes, scan logs, tamper alerts, and curated activity events

4.1.4 Scope Statement

The Secure QR Code system addresses the inherent insecurity of publicly displayed QR codes, which can be physically replaced or tampered with by malicious actors. The system will deliver a full-stack authenticated QR code platform consisting of a backend API server, two web-based surfaces, and a mobile application. Reaching a QR code's destination requires the dedicated mobile application; users scanning with a standard camera still receive a full server-side verification result in the browser, but are never given the destination URL.

What will be INCLUDED:

QR code generation with cryptographically signed JWT tokens, configurable expiry, and an optional descriptive label. Generation is restricted to authenticated administrators.

A server-side verification system that validates the token signature and expiry and checks the code's status before any database lookup is trusted.

Two strictly separated JWT types — QR tokens embedded in printed codes and authentication tokens for user sessions — signed with different secrets that are never interchanged, plus a per-account token version counter that revokes all existing sessions on password change, password reset, or role change.

An app-required access model: the destination URL is delivered only through the mobile application. Browser users receive the full safe/warning verdict plus an application download prompt.

Backward compatibility with all standard QR code scanners — any device can scan the generated codes without special software and will reach a real, working page.

Two web surfaces serving distinct purposes:

QR Code Landing Page: shown to regular scanner users when a QR code is scanned. Performs server-side token validation, displays a verification result, and prompts the user to download the dedicated app.

Admin Management Dashboard: a full web application for QR code management (generate, activate, blacklist), QR code display and printing after generation, alert review with photo and GPS evidence, user administration, account settings, and scan metrics visualisation.

Channel-appropriate delivery for app users and regular scanner users: the public verification endpoint serves standard scanners unchanged, while a dedicated authenticated endpoint serves the mobile application and attributes each scan to the signed-in account. The User-Agent string is recorded as telemetry only and is never used as an authorisation mechanism.

A four-state QR code lifecycle (active, suspicious, blacklisted, expired) with enforced integrity rules: time-expired codes are swept to the expired state on administrator reads, expiry is terminal because it is baked into the printed token, and a user tamper report may promote a code only from active to suspicious.

A blacklist management system for flagging and deactivating compromised QR codes, taking effect on the very next scan.

A tamper alert system allowing signed-in mobile app users to report suspected tampering with GPS location, photo evidence (image types only, 5 MB limit, stored in cloud object storage with only the URL held in the database), description, and optional contact details.

Scan telemetry logging including timestamp, IP address, User-Agent, and validation result — recorded for every scan attempt, including failed and forged-token scans. Optional GPS is recorded only for tamper reports.

A complete user authentication system shared by the web dashboard and the mobile application: registration, login, forgot-password and reset-password using single-use expiring tokens, and logged-in password change. Passwords are hashed with bcrypt.

User administration for administrators: user listing with search and pagination, role promotion and demotion protected by a last-administrator guard and a self-role-change block, administrator-created accounts, and user deletion. The first administrator account is seeded from environment variables and never hardcoded.

Server-side per-user scan history, so an app user’s history survives reinstallation and device changes.

Operational reporting features: CSV export of the QR code register, a curated recent-activity feed of meaningful system events, dashboard metric cards, and period-over-period metric comparison.

What will NOT be included:

Extension or modification of the ISO/IEC 18004 QR code standard (listed as an optional extra — outside current scope).

Time-based redirect service (listed as an optional extra — outside current scope).

Integration with third-party enterprise systems or external APIs beyond the core platform.

A browser path to the destination URL. Regular scanner users receive verification and an application download prompt only; the destination is deliberately never passed to the browser.

Anonymous tamper reporting and anonymous destination access. Reporting and destination access require a signed-in account, although the browser landing-page verification path remains entirely account-free for the public.

A standalone QR Code Display web page. This function is delivered inside the authenticated admin dashboard instead, because it only ever serves administrators.

Native iOS app distribution via the Apple App Store (out of scope for this phase; Android-first development). Under the app-required model, iOS users can see the verification result but cannot reach the destination URL.

Hardware or physical security of printed QR codes — the system cannot prevent physical replacement of codes, only detect and respond to it.

A server-side allowlist of permitted destination domains, and a manual location-entry fallback when GPS permission is denied. Both are re-scoped as future work; the allowlist risk is low because generation is administrator-only and GPS remains optional.

Account suspend and restore, two-factor authentication, email notification preferences, administrator-initiated password resets for other users, and real email delivery for password resets (demonstration mode returns the reset link in the API response).

Migration of authentication to a managed identity provider — the custom JWT implementation is complete and QR tokens cannot be issued by such a provider.

Automated downtime monitoring and scan-volume anomaly detection. Database backups are delegated to the managed database provider, and the recent-activity feed reports only real recorded events.

4.2 Work Breakdown Structure Summary

1. Project Management

1.1 Develop and maintain Project Management Plan

1.2 Conduct and document client meetings

1.3 Manage team task allocation and progress tracking

1.4 Conduct risk identification and mitigation planning

1.5 Prepare and submit milestone deliverables

1.6 Maintain the Design Change Register and obtain client approval for behaviour-changing modifications (notably the app-required access model)

2. Requirements and Analysis

2.1 Gather and document functional requirements from client brief

2.2 Identify and document non-functional requirements

2.3 Clarify open items with client (Peter and Mike)

2.4 Produce and submit Requirements and Analysis Document

2.5 Obtain client sign-off on requirements

2.6 Amend the actor definitions for the introduction of user accounts, and re-scope the destination allowlist and GPS manual-entry fallback requirements

3. System Architecture and Design

3.1 Define overall system architecture (backend, web surfaces, mobile app)

3.2 Design database schema (users, QR codes, scan logs, alerts, activity log)

3.3 Define API endpoint structure and contracts

3.4 Define token signing and validation approach (separate QR and authentication JWT secrets, plus session revocation)

3.5 Define the channel identification approach for app and browser users, and the authenticated endpoint used by the app

3.6 Define the two web surface flows and their purposes

3.7 Produce Architecture Design Document

4. Backend Development

4.1 Set up Node.js + Express project and managed PostgreSQL database

4.2 Implement database schema using Prisma ORM

4.3 Implement QR code generation endpoint (POST /api/qr/generate), restricted to administrators, accepting destination URL, expiry, and optional label

4.4 Implement token signing service using JWT, with separate QR and authentication secrets

4.5 Implement QR verification endpoints

4.6 Implement alert submission endpoint (POST /api/alert/report), requiring authentication

4.7 Implement authentication endpoints and access-control middleware

4.8 Implement admin API endpoints

4.9 Implement per-user scan history endpoint for the mobile application

5. Web Page Development

5.1 QR Code Landing Page (for regular scanner users)

5.2 QR Code Display function (integrated into the admin dashboard)

5.3 Admin Management Dashboard

6. Mobile Application Development

6.1 Set up React Native project

6.2 Implement login and registration screens, with registration confirming account creation and requiring an explicit sign-in

6.3 Implement QR code scanner screen

6.4 Implement token submission to the authenticated backend verification endpoint

6.5 Implement verification result screen (safe / warning) and destination access for valid codes

6.6 Implement tamper alert reporting screen

6.7 Implement personal scan history screen backed by the server

6.8 Set custom User-Agent header (SecureQRApp/1.0) on all requests for channel telemetry

6.9 Build and publish the Android APK and configure the download link used by the landing page

7. Testing

7.1 Develop test plan covering all functional requirements, re-baselined for the app-required access model

7.2 Unit testing of backend services (token, QR generation, alert)

7.3 Integration testing of all API endpoints, including authentication and role-based access control

7.4 End-to-end testing of the full scan and alert flow

7.5 Backward compatibility testing with standard QR scanners

7.6 Web page testing across mobile and desktop browsers

7.7 User acceptance testing with client

7.8 Document and report test results

8. Documentation

8.1 Write and maintain Requirements and Analysis Document

8.2 Write Architecture Design Document

8.3 Write User Manual for admin dashboard and mobile app, including the user administration chapter

8.4 Write Installation and Deployment Guide covering database connection rules, build and deployment commands, and file storage configuration

8.5 Maintain the Design Change Register and record the known limitations of the delivered system

8.6 Produce Final Project Report

5. Project Time Management

The work breakdown structure is developed for this project and included in Appendix B. It describes all activities necessary for finishing the project. About 65 separate activities are divided into eight activity categories: Project Management, Requirements and Analysis, System Architecture and Design, Backend Development, Web Page Development, Mobile Application Development, Testing, and Documentation.

All the activities are associated with durations measured in hours or days. Durations were determined by considering team members' experience in executing similar coding activities and difficulty levels of activities. Thus, the task to configure the backend project using Node.js and Express takes a day since team members have already accomplished this task during previous units. The activity to develop a complete admin dashboard including all web pages and API calls will take five days as there are many activities involved.

Dependencies exist between different tasks. In cases where Task B can only start after Task A is finished, a dependency exists. Some of the dependencies that have been identified in this QR code generation project include:

- QR Code landing page generation can only happen after the QR verification endpoint has been created.
- Mobile application scanning screen development can only happen after the QR verification endpoint has been created.
- Testing of tamper alert system can only begin after alert submission endpoint has been created.
- Compilation of user manuals can only be undertaken when the project is near completion because screens and buttons must be consistent with the final system.
- Critical path is that path along which all delays will affect the total project duration; therefore, delay in the execution of any task will affect the project deadline of 1 August 2026.
- The critical path in this project includes:
  Client signs off on the requirements → Design database schema → Token signing service → QR verification endpoint → Mobile scanner screen development → Alert submission endpoint → Testing of final system

If there is any delay with any of the above tasks, prompt action must be taken.

5.1 Project Milestone Dates

This assignment started on 20 May 2026. Every assignment will have to be done by the time indicated below. Deadline will be strictly observed for all deliveries. Any extension will only be granted if there is an emergency situation.

Deliverable	Due Date
Team Charter submission	23 May 2026
Project Management Plan V1.0	6 June 2026
Requirements and Analysis Document	8 June 2026
Design Document	27 June 2026
Self and Peer Evaluation 1 (SPE1)	27 June 2026
Final Project Submission (all code and all documents)	1 August 2026
Project Presentation PowerPoint file	1 August 2026
Demo media file	1 August 2026
Promo Video file	1 August 2026
Self and Peer Evaluation 2 (SPE2)	1 August 2026

5.2 Timeline for Development Work

Tasks for requirements finalization, database design and backend configuration will take place during the second sprint from June 6 to June 19, 2026. These tasks will form the basis for further development in the third sprint. It is worth noting that even with this change to the timeline, the requirement final submission date 1st August still remains realistic.

5.3 Gantt Chart

To help us track the entire project timelines, a simple Gantt chart has been made. In a Gantt chart, each task identified from the Work Breakdown Schedule appears on the Gantt chart in form of a horizontal bar on the calendar. The point where the bar starts denotes the starting date while the end of the bar denotes the ending of the task.

The Gantt chart's timeline covers the period from 6th May 2026 to 1st August 2026.

The Gantt chart can be found in Appendix B of this document.

5.4 Network Diagram

A network diagram has been created to help better understand the relationships between tasks. In this network diagram, rectangles are used for tasks and arrows are used to indicate the task that should be completed before a new task begins.

From the network diagram, one realizes that the most crucial task in the whole project is the QR verification endpoint because it depends on five other tasks: landing page, mobile scanner, the process of redirection, the scan log capability, and the testing process.

The Network Diagram can be found in Appendix B of this document.

5.5 Weekly Progress Monitoring

Patricia, the project manager, will analyze the actual performance against the expected performance at 9:00AM every Monday morning based on the Gantt chart. Progress reporting will take place weekly via the communication channel of the MsTeams.

In case the project team falls behind schedule for more than two days, the conflict resolution procedure specified in Section 9 is employed. It includes the following steps:

1. The person in charge of the corresponding activity provides an explanation for the delay.

2. Team analyzes the problem from all angles and discusses it.

3. Team explores possible courses of action and agrees upon a corrective course of action. It involves either the additional work done by the person responsible for that activity or streamlining the activity by excluding unnecessary components or delaying another activity.

4. Outcome of the discussion is registered in the minutes.

The delays are always discussed and resolved immediately.

5.6 Buffer Days Prior to Deadlines

Since all deliverables are handed in strictly according to the set deadlines, buffer days have been added prior to each deadline. It means that all deliverables should be fully prepared two days prior to the deadline.

For instance, the deadline for the Final Project Submission is 1 August. As a result, the completion of all the code, writing all necessary documentation, and testing should take place by 30 July. Two days left will be the buffer period for addressing possible problems. In case any problem emerges in the course of work on 30 July, one can still fix them within the time period until 1 August.

Concerning the 1 August deadline, the period of five buffer days from 27 July until 31 July has been selected. This period of five days is intended solely for solving emergency problems that may emerge in the process of preparation of the deliverables. Any additional work is not expected within this timeframe.

5.7 Recording Actual Finish Dates Against Planned Finish Dates

An estimation comparison spreadsheet has been designed in Excel sheets where there are five columns namely Task ID, Task name, Planned finish date, Actual finish date, and days variance.

Whenever a particular task is completed by its assignee, the assignee will then enters the actual date in the relevant cell and the project manager calculates the variance. If the actual completion date and the planned completion date coincide, then the variance would be zero while in case of a two-day delay, variance would be +2.

Once two sprints are done, the variance column is verified for the precision of the estimate. If there is a significant amount of variance in the schedule, then the estimate and prioritization of tasks will be evaluated and modified if required.

The Gantt chart is reviewed and updated every Thursday after the weekly meeting of the team and the new version of it is named with the new date in it. The older versions will be kept in a separate folder.

6. Project Quality Management

Quality is described as the level to which a system fulfills the specified criteria and achieves its intended function. In this project:

- **Adherence to requirements:** Ensuring that the secure QR system fulfills the functional and security criteria, including QR, expiration verification and safe redirection.

- **Usability and effectiveness:** Guarantee that the system is functional, dependable and accessible for users on mobile.

6.1 Component quality checklist

Component	**Checklist**	
QR code generator	Creates unique UUID, scannable QR, free of duplicates	
QR validation API	Accurately checks QR and rejects expired or blacklisted codes.	
Mobile application	Functional QR scanning, QR verification and tamper report submission	
Redirect system	Redirects to the URL within ≤ 3 seconds	
Database	Data is stored without any corruptions with backups and recovery support.	
Reporting system	Accurately records GPS, description, photo evidence and submission timestamp.	
Admin dashboard	Correctly display the right alerts, system status information and scan metrics.	

6.2 Quality assurance testing

- Perform testing to check on the accessibility of the mobile application.

- Perform testing on the QR code with and without the mobile application to check if the redirect system is working.

- Perform testing to check if the download link for the mobile application is working.

- Perform testing to check if QR code has an expiry

- Conduct tests to ensure that the blacklisted QR codes are accurately identified and blocked from redirecting users.

- Conduct tests to verify whether alerts are correctly submitted and is visible in admin dashboard.

6.3 Performance metrics, integrity safeguards and acceptance criteria

QR code generator

Metrics:

- QR code generation success rate ≥ 99%
- Duplication rate of UUID = 0%

Acceptance criteria:

- Every single QR code generated is unique
- No duplicated QR codes in the database
- QR codes can be scanned successfully using the standard phone camera.

QR validation API:

Metrics:

- QR code expiry rate = 100%
- Blacklisted QR code detection rate = 100%

Acceptance criteria:

- QR code after the expiry period and invalid should be shown as invalid.
- Blacklisted QR codes must be blocked from redirecting the users.

Mobile application:

Metrics:

- Successfully QR code scan rate ≥ 99%
- Tamper report submission success rate ≥ 95%
- Crash/downtime rate ≤ 0.1%
- Average delay time < 3 seconds

Acceptance criteria:

- Users must be able to scan the QR code successfully
- Reports must be submitted successfully
- Alerts must be displayed correctly with accurate details (GPS, image, description, time)

Redirect system:

Metrics:

- Redirection of URL time ≤ 3 seconds

Acceptance criteria:

- Time taken for the redirection of URL to take place should be 3 seconds or less.

Database:

Metrics:

- Corrupted data in the database = 0%

Acceptance criteria:

- Any data stored in the database must be inherently uncorrupted

Reporting system:

Metrics:

- Report submission rate = > 95%
- Accuracy of GPS rate > 95%
- Successful image upload rate > 95%

Acceptance criteria:

- All reports must be safely stored and can be retrieved at any point of time
- Reports must include accurate GPS location, description, timestamp and images

Admin dashboard:

Metrics:

- Accuracy of alerts shown to admin ≥ 99%

Acceptance criteria:

- Alert should contain description of device or location using mobile device information
- If the user wishes to share; reporter name and contact information
- Photo of QR code and surroundings

7. Project Communications Management

The project communication management guidelines outlined how project information was created, shared, stored, and reviewed throughout the secure QR code development project. Effective communication was crucial for the project's successful completion, as it involved clients, academic mentors, project managers, developers, testers, business analysts, and members responsible for project documentation.

Our project team utilized WhatsApp, Microsoft Teams, email, OneDrive, GitHub, and face-to-face meetings for various purposes. These communication tools were used for requirements confirmation, task allocation, development coordination, document sharing, version control, testing, and final presentation preparation.

During the project, our team encountered some communication issues, primarily including delayed responses from some members and team members not promptly informing others of their completed modifications. To address these issues, our team increased meetings and group chat communication, reconfirmed each person's responsibilities, used GitHub to manage different versions of the source code, created a separate "Landing Page Version 2" to avoid overwriting existing modifications, and conducted a centralized system integration and review before the final presentation.

7.1 Stakeholder Analysis

The following stakeholders are involved in the project communication process:

Stakeholder	Role in Project	Communication Needs	Communication Method	Frequency
Peter Cole and Mike Groeneweg	Clients	Provide project expectations, confirm requirements, review milestones, provide feedback	Email, Microsoft Teams	After major milestones or when client clarification is required
Academic Supervisor	University project supervisor	Provide academic guidance, monitor team progress, clarify unit expectations	Microsoft Teams, Email, weekly meeting	Weekly or when required
Patricia Orenza	Project Manager and main contact person	Coordinate the team, assign work, track progress,	WhatsApp, Microsoft Teams, Email, OneDrive	Regularly throughout the project

		communicate with clients and supervisor		
Lim Yu Jun Felixia	Business Analyst and documentation contributor	Clarify requirements, manage user stories and contribute to project documentation	WhatsApp, OneDrive, Microsoft Teams	Weekly and before major submissions
Roy Yoo Hojin	Back-end Developer and System Integration Developer	Develop the back-end functions, connect the front-end components to the back-end system, communicate technical changes and support system integration	WhatsApp, GitHub, Microsoft Teams, OneDrive	Several times per week during development and integration
Gu Qiyu	Front-end Developer	Develop the Mobile App, Admin Web and Landing Web front-end components and coordinate code integration	WhatsApp, GitHub, OneDrive, Microsoft Teams	Several times per week during development and integration
Vanessa Tan	Tester	Test the system components, record test results, identify defects and communicate the corrections required	WhatsApp, GitHub, OneDrive	During testing and before milestone submissions
Alvin Rui Long Yeow	Tester	Test the system components,	WhatsApp, GitHub, OneDrive	During testing and

		assist with identifying defects and communicate testing feedback to the developers		before milestone submissions
Whole Project Team	Project delivery team	Share updates, discuss problems, review tasks, complete deliverables	WhatsApp, Microsoft Teams, OneDrive, GitHub	Weekly meetings and ongoing messages

The main external stakeholders are Peter Cole and Mike Groeneweg, who are identified as the clients for the Secure QR Code Development project. The project objective is to build a secure QR code authentication and management system that supports expiry timeout, unique authentication codes, redirects, telemetry, alerts, and mobile scanning. The Team Charter also identifies Patricia Orenza as the Project Manager and main point of contact for the clients.

7.2 Communication Methods and Frequency

We will use different communication methods depending on the purpose and urgency of the information.

Communication Activity	Purpose	Method Used	Frequency	Responsible Participants
Daily team communication	Quick questions, reminders, progress updates and issue discussions	WhatsApp	As required and checked regularly	All team members
Team meetings	Review progress, discuss problems, reconfirm responsibilities and determine the next actions	Microsoft Teams or face-to-face meetings	Weekly or more frequently before major deadlines	All team members
Client	Clarify	Email,	At major	Patricia

communication	requirements, provide milestone updates and obtain feedback	Microsoft Teams	milestones or when clarification was needed	Orenza and relevant members
Supervisor communication	Obtain academic guidance, review project progress and clarify submission requirements	Microsoft Teams and Email	Weekly or when needed	All team members
Document sharing	Store and share project documents, meeting minutes, plans, and reports	OneDrive	Updated whenever documents change	Lim Yu Jun Felixia
Code collaboration	Store source code, track code changes, review development progress	GitHub	Throughout development and integration	Roy Yoo Hojin and Gu Qiyu
Testing communication	Report bugs, testing results, and fixes required	WhatsApp, GitHub, OneDrive，Microsoft Teams	During testing and before the final demonstration	Alvin Rui Long Yeow and Vanessa Tan
Final integration communication	Review component readiness, resolve outstanding issues and prepare the demonstration	WhatsApp, GitHub and face-to-face meetings	Intensively before the final demonstration	Whole project team

WhatsApp is our primary channel for rapid communication within the team. Microsoft Teams, email, and in-person meetings are used for more formal discussions and communication with stakeholders. OneDrive is used to store and share project documents, while GitHub is used to manage source code and different development versions.

7.3 Response Time and Communication Rules

To ensure that communication is effective and that project delays are avoided, the team will follow the response rules below:

Communication Situation	Expected Response Time
General WhatsApp message	Within 24 hours
Urgent project issue or deadline-related message	Within 6 hours
Client or supervisor email	Within 24–48 hours
Document review request	Within 2 working days
GitHub pull request or code review	Within 1–2 working days
Bug report during testing period	Within 24 hours

Team members are required to check WhatsApp regularly and OneDrive before meetings and document submissions. Developers also need to frequently check GitHub during development, testing, and integration.

If a team member is unable to complete a task on time, the project manager should be notified as soon as possible. This allows the team to adjust schedules, clarify responsibilities, or reassign tasks as necessary.

However, in practice, some team members fail to respond promptly, sometimes even making changes without clearly informing other team members. These issues necessitate further communication and discussion before integration and submission.

7.4 Tone of Communication

All project communication must be professional, respectful and clear. We need to be mindful of this because our project involves working with staff, clients and team members from different technical backgrounds. So, we will use a constructive tone in all our communication.

When we communicate within our team we can use direct language. We should still be respectful and focused on the task at hand. However, when we communicate with clients and supervisors we should use language.

Here are some guidelines for writing emails:

- Use a subject line
- Start with a greeting
- State the purpose of the message
- Specify the action
- Include a deadline if applicable
- End with a closing

Sometimes disagreements will happen. When they do, we will focus on the project issue not on criticism. We will discuss problems based on evidence project requirements and task priorities. If we can't resolve a disagreement, within our team the Project Manager will escalate the issue to the supervisor.

7.5 Documentation Formatting

Our project documentation was primarily written in Microsoft Word and stored in a shared OneDrive folder. The team used numbered headings, tables, charts, and screenshots to organize information. Important tables and images were placed near explanatory text.

Before final submission, our team checked the following aspects of the documentation:

- Spelling and grammar
- Consistency of headings and formatting
- Correct page numbers
- Correct document version
- Completeness of necessary sections
- Compliance with evaluation guidelines and template requirements

To minimize duplicate content and version conflicts, team members should ensure they are using the latest version before modifying any files.

7.6 Technologies and Access Methods

The project will use the following technologies and access methods for communication, document management, and development:

Tool / Technology	Purpose	Access Method
WhatsApp	Fast team communication, reminders, informal updates	Team group chat
Microsoft Teams	Formal meetings, supervisor meetings, client meetings, online discussions	Team meeting links or invited channels
Email	Formal communication with clients, supervisor, and important approvals	University email accounts
OneDrive	Shared document storage, meeting minutes, reports, planning documents	Shared team folder
GitHub	Source code storage, version control, development collaboration, issue tracking	Shared GitHub repository
Microsoft Word / PDF	Project documentation and submission files	Stored in OneDrive
Excel	Progress tracking and task management	Stored in OneDrive or shared workspace

We only allow team members to use OneDrive and GitHub. They are only allowed access when clients genuinely need it, such as when we want them to see our current work progress, project status, or agreements reached. We take great care to safeguard passwords, API keys, database credentials, and private tokens. We do not share sensitive information such as passwords, API keys, database credentials, and private tokens on OneDrive and GitHub in WhatsApp or other public documents.

7.7 Communication Issues and Corrective Actions

During the project, our team encountered two main communication issues.

First, some team members did not respond to project messages promptly. This increased the difficulty of confirming information, clarifying responsibilities, and completing related tasks. Second, some modifications were not clearly explained to the team before or immediately after completion. This led to other members continuing to use older versions or unknowingly overwriting existing modifications.

To address these issues, our team took the following measures:

1. When daily communication was insufficient, our team increased meetings and group chats.
2. We reconfirmed member responsibilities to clarify the responsibilities for development, testing, project documentation, and system integration.
3. We used GitHub to store the project source code and manage different versions.
4. When the existing Landing Web code had been modified by other members, the team created a separate folder named "Landing Page Version 2". This prevented the new front-end code from overwriting other members' modifications while maintaining two versions for comparison and integration.
5. Before the final demo, our team conducted a centralized check and integration to confirm that the Mobile App, Admin Web, Landing Web, and other supporting components were ready for demonstration.

These measures reduce the risk of losing work results and help the team complete the final system integration and demonstration preparation.

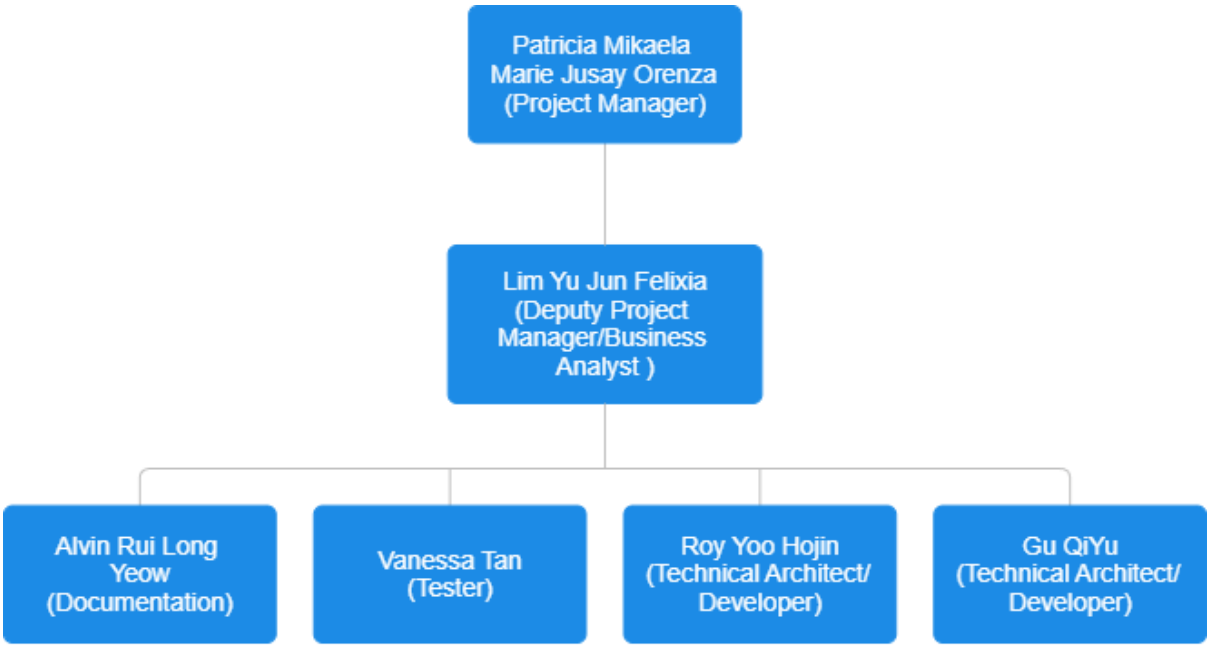
7.8 Final Communication Performance Review

Overall, communication methods supported project completion, although we weren't always efficient at every stage. WhatsApp enabled rapid communication, OneDrive provided document sharing, and GitHub offered crucial mechanisms for code storage and version control. Microsoft Teams, email, and in-person meetings were used for formal discussions, stakeholder communication, and final coordination.

Despite some communication difficulties encountered during the project, our team strengthened coordination, reviewed project components, and completed necessary integration and commit preparations before the final presentation

8. Project Human Resources Management

Organisational Chart:



Responsibility Assignment Matrix:

		Resource					
	Tasks	Patricia	Felixia	Alvin	Vanessa	Roy	Qiyu
Project Management	Develop Project Management Plan	A/R	A	C	C	C	C
	Conduct client meetings	A/R	A/R	R	R	R	R
	First client meeting	A/R	A/R	R	R	R	R
	Second client meeting	A/R	A/R	A/R	A/R	A/R	A/R
	Client walk-through / UAT	A/R	A/R	A/R	A/R	A/R	A/R
	Task allocation and tracking	A/R	A	I	I	I	I
	Risk identification and mitigation	A	A	I	R	I	I
	Milestone deliverable preparation	A/R	A	C	C	C	C
Requirements and	Gather functional	A	A/R				

	requirements						
	Gather non-functional requirements	A	A/R				
	Clarify open items with client	A	A/R	C	C	C	C
	QR code expiry duration	A	A/R			I	I
Analysis	User information fields	A	A/R			I	I
	Expected daily user volume	A	A/R			I	I
	Telemetry requirements	A	A/R			I	I
	Log retention period	A	A/R			I	I
	Produce R&A Document	A/R	A/R	R	R	R	R
	Obtain client sign-off	A/R	A	I	I	I	I
	Define overall system architecture					A/R	A/R
	Design database schema					A/R	A/R
	qr_codes table					A/R	A/R
	users table					A/R	A/R
	scan_logs table					A/R	A/R
	alerts table					A/R	A/R
System Architecture and Design	Define API endpoint structure					A/R	A/R
	Define token signing approach					A/R	A/R
	Define user-agent detection logic					A/R	A/R
	Define three web page flows					A/R	A/R
	Produce Architecture Design Document					A/R	A/R
Backend Development	Project and environment setup					A/R	A/R
	Database setup					A/R	A/R

	QR code generation endpoint					A/R	A/R
	JWT token signing					A/R	A/R
	QR image generation					A/R	A/R
	Database record creation					A/R	A/R
	Token service					A/R	A/R
	QR verification endpoint					A/R	A/R
	Token signature validation					A/R	A/R
	Token expiry check					A/R	A/R
	Blacklist and whitelist check					A/R	A/R
	User-agent detection and branching					A/R	A/R
	Scan logging					A/R	A/R
	Alert submission endpoint					A/R	A/R
	Photo upload handling					A/R	A/R
	GPS and form data storage					A/R	A/R
	Automatic status update					A/R	A/R
	Admin API endpoints					A/R	A/R
	GET /api/admin/qrcodes					A/R	A/R
	GET /api/admin/qrcodes/:id					A/R	A/R
	PATCH /api/admin/qrcodes/:id					A/R	A/R
	GET /api/admin/alerts					A/R	A/R
	GET /api/admin/metrics					A/R	A/R
Web Page Development	QR Code Landing Page					A/R	A/R
	Server-side token					A/R	A/R

	validation						
	Safe result display					A/R	A/R
	Warning result display					A/R	A/R
	App download prompt					A/R	A/R
	Mobile responsive design					A/R	A/R
	QR Code Display Page					A/R	A/R
	QR code image display					A/R	A/R
	QR code details display					A/R	A/R
	Download option					A/R	A/R
	Print option					A/R	A/R
	Admin Management Dashboard					A/R	A/R
	Project and environment setup					A/R	A/R
	Application layout and navigation					A/R	A/R
	Dashboard metrics page					A/R	A/R
	QR Code Management page					A/R	A/R
	Generate QR code form					A/R	A/R
	QR codes listing table					A/R	A/R
	Blacklist and activate actions					A/R	A/R
	Alerts review page					A/R	A/R
	API integration					A/R	A/R
Mobile Application	Project and environment setup					A/R	A/R
	QR code scanner screen					A/R	A/R
	Token submission to backend					A/R	A/R
	Verification result					A/R	A/R

	screen						
	Tamper alert reporting screen					A/R	A/R
	Automatic alert trigger					A/R	A/R
	Manual alert trigger					A/R	A/R
	GPS location capture					A/R	A/R
	In-app photo capture					A/R	A/R
	Description and contact form					A/R	A/R
	Alert submission to backend					A/R	A/R
	Custom User-Agent header					A/R	A/R
	App permissions configuration					A/R	A/R
Testing	Develop test plan			R	A/R		
	Backend unit testing			R	A/R	C	C
	Token generation tests			R	A/R	C	C
	Token validation tests			R	A/R	C	C
	QR generation tests			R	A/R	C	C
	API integration testing			R	A/R	C	C
	POST /api/qr/generate tests			R	A/R	C	C
	GET /api/qr/verify tests			R	A/R	C	C
	POST /api/alert/report tests			R	A/R	C	C
	Admin endpoint tests			R	A/R	C	C
	End-to-end flow testing			R	A/R	C	C
	Backward compatibility testing			R	A/R	C	C
	Web page testing			R	A/R	C	C
	Landing page tests			R	A/R	C	C
	QR display page tests			R	A/R	C	C
	Dashboard tests			R	A/R	C	C
	Mobile app testing			R	A/R	C	C

	User acceptance testing	A/R	A/R	I	I	I	I
	Document and report test results	A	R	R	A/R	C	C
Documentation	Requirements and Analysis Document		A/R	A/R			
	Architecture Design Document			A/R		C	C
	Project Management Plan	A/R	R	A/R	R	R	R
	Admin dashboard user manual			A/R		C	C
	Mobile app user manual			A/R		C	C
	Installation and Deployment Guide			A/R		C	C
	Final Project Report	A/R	A/R	A/R	A/R	A/R	A/R
Key:							
R: Responsible. They are people who put in the work to complete the tasks.							
A: Accountable. People who are answerable for the task's completion by delegating the work and review to ensure it meets standards							
C: Consulted. People who provide input based on their expertise							
I: Informed. People who are informed or kept up to date on the progress of the task at hand							

Our team will have weekly online meetings on Thursday or Friday nights to follow up on any tasks given and to keep track of each other’s progress. Any other questions that we have can be asked and answered via our group chats or through this meeting.

To resolve any conflicts in the team, we will lay out the situation at hand and the perspectives of both sides of the situation. As a team, we will determine who is at fault and what compromises both sides should make.

9. Project Risk Management

Risk management involves the identification, evaluation, monitoring and response to occurrences that could negatively affect the project goals. The subsequent risk register outlines possible risks linked to the Secure QR Code Authentication System project, the probability and impact, the mitigation strategies and the members who are responsible, these risks will be constantly reviewed during our weekly meetings and changes will be made as necessary throughout the entire project duration.

9.1 Risk register

Risk ID	Risk name	Description	Risk category	Impact	Probability	Mitigation/ response strategy	Person responsible
R01	Scope creep	Extra requirements or requests of features that could potentially extend the project beyond the existing resources and time.	Project management	High	High	Prioritize the essential deliverables, uphold a clear project scope and categorize non-essential features as potential upgrades.	Patricia
R02	User refusing device permissions	Users might not allow permissions for the camera, storage or GPS which is required for QR scanning and the reporting of incidents.	Technical	High	Medium	Present explicit permission instructions within the application itself and offer manual reporting choices.	Roy, Qiyu
R03	Data leak or information exposure	Confidential information like location details, user personal information or incident logs	Security	High	Low	Encryption of sensitive data stored in the database, mandating safe	Vanessa, Alvin

		could be exposed to someone with unauthorized access.				communications using HTTPS and establish role-based access control.	
R04	QR Code replication attack	Authorized QR codes can be replicated and deployed in unauthorized places which misleads the users.	Security	High	Medium	Encourage and enable users to flag any suspicious QR code and implement anomaly identification based on scan location to detect any unusual geographic patterns and potential QR replication attempts.	Vanessa, Alvin
R05	Issues with mobile	Applications might function differently	Technical	Medium	Low	Using various android versions and	Roy, Qiyu

	device compatibility	depending on the OS versions or device models.				devices during the development process while conducting tests.	
R06	Backend server downtime	Any server failures can hinder the operation of QR validation, authentication or redirection.	Infrastructure	High	Medium	Utilize reliable hosting with automated daily backups, implementing real-time monitoring including alerts that will trigger if downtime exceeds 3 minutes and ensuring to come up with a well-defined disaster recovery plan including service restart and	Roy, Qiyu

						database restoration.	
R07	Data loss or database malfunction	Records of QR, telemetry information and user submissions might be lost because of any corruption within the database or system malfunction.	Technical	High	Low	Set up automated daily backups of the database, database replication for extra security and establishing disaster recovery protocols to retrieve telemetry logs, QR data and user submissions during system downtime or database corruption.	Roy, Qiyu
R08	Concerns regarding APK	Users might be hesitant to install the application	Business	High	Medium	Include precise installation	Felixia, Patricia

	sideloading trust	because of Android security alerts when activating “Install from Unknown Sources” leading to decreased overall adoption of the system.				guidelines with detailed steps with visual aids on the website to assist users in installation successfully.	
R09	Incorrect QR classification	Valid QR codes might be mistakenly classified as harmful or suspicious.	Operational	High	Low	Set up validation guidelines, offer administrator evaluation processes and perform tests.	Vanessa, Alvin
R10	Low user adoption	Users might choose to use the standard built in phone camera scanners rather than downloading the secure QR application.	Business	High	Medium	Implement browser-based verification options and have clear and concise download instructions.	Felixia, Patricia

R11	Delay in project schedule	Development task might take longer than the expected duration causing a delay in the project milestones and final delivery.	Project management	High	High	Ensure to have regular progress checks, effectively allocate tasks and prioritise critical functionality.	Patricia
R12	SSL/TLS certificate expiry or misconfiguration (Security)	If the app depends on a particular certificate for communicating with the Node.js backend, if the certificate expires or is not updated properly, it can cause the application to not be able to connect to the server and cease functioning correctly.	Security	High	Medium	Establish effective certificate lifecycle management and monitoring and guarantee that certificate renewal procedures are organized and communicated ahead of time to avoid any service interruptions.	Roy, Qiyu (primary) + Vanessa, Alvin (security)

R13	GitHub merge conflicts	Work may be accidentally overwritten, or merge conflicts may occur due to concurrent code modification.	Technical	Medium	Medium	Utilize feature branches, code assessments, pull requests or regular commits.	Roy, Qiyu
R14	Malicious file upload	During submission of tampering report, attackers may include malicious files that are disguised as images.	Security	High	Medium	Restrict file types to approve only image formats, enforce size limits for files and validate MIME types.	Vanessa, Alvin
R15	Misunderstanding of requirement	Team may misunderstand what the requirements of the client are which can lead to the need to rework, changes to completed work or delays.	Requirements	High	Medium	Perform routine reviews, have regular showcase of progress and secure approval to ensure that everyone is	Patricia

						on the same page and can get their questions across if any.	
R16	Unavailability of member	A member might become unavailable due to health issues, academic responsibilities or personal obligations.	Project management	High	Medium	Make sure that documentation is up to date, reassign task if needed, help each other out.	Patricia
R17	Increased system acceptance and stakeholder engagement	The Secure QR Code system experiences greater-than-anticipated adoption from both the businesses and users driven by significant interest in QR security resulting in	Business / Opportunity	Medium	Medium	Design system architecture for scalability, confirm backend can accommodate higher demand and frequently monitor usage	Patricia

		enhanced feedback, utilization and potential scalability requirements that exceed initial projections.				metrics. Make sure to be prepared for possible performance improvement or feature enhancement if usage surpasses expectations.	
R18	Replacement of physical QR code.	Attackers could directly replace the authentic QR code by sticking a malicious sticker over it.	Security	High	High	QR code scanned will be considered invalid on our application since it is not recognized in our database. Additionally, our tampering report function enables users to quickly report it to us	Roy, Qiyu, Vanessa, Alvin

						so we can blacklist it immediately.	

9.2 Probability/impact matrix

IMPACT

PROBABILITY		**Low**	**Medium**	**High**
	Low			R01, R11
	Medium		R13, R17	R02, R04, R06, R08, R10, R12, R14, R15, R16
	High		R05	R03, R07, R09, R18

Probability:
Low = Unlikely to happen
Medium = can happen throughout project duration
High = likely to happen

Impact:
Low = slight delay or no significant impact
Medium = influences module or sprint
High = threatens project completion or scope

10. Project Cost Management

This project cost management section explains how we estimate, monitor, and control the human, material, and technical resources required for the secure QR code development project. Because this is an academic project, our team members do not receive salaries, and we have not purchased large equipment. However, the absence of direct payments does not mean that the resources used in our project are without cost.

Therefore, we adopted an input-based costing approach. We measure human resource costs in "person-hours" rather than converting working hours into wages. For material and technical resources, we measure them using units such as the number of devices, device usage hours, number of user accounts, usage weeks, and project-related travel.

This approach more accurately reflects the actual resources used in our project compared to recording every cost item as zero.

10.1 Cost Estimation Method

We used an effort-based method to estimate and measure the cost of our project. Human-resource effort was measured in person-hours instead of being converted into monetary salaries. Our physical and technical resources were measured using quantities such as the number of devices, duration of access and frequency of use.

Our human-resource records covered 13 weekly reporting periods from 6 May 2026 to 1 August 2026. For each week, we recorded the planned and actual hours contributed by Patricia, Felixia, Vanessa, Alvin, Roy and Qiyu. These hours included project management, requirement analysis, development, testing, documentation, meetings, system integration and preparation for the final submission.

We compared each member’s actual hours with their planned hours to identify any differences in resource use. A positive variance indicates that more hours were required than planned, while a negative variance indicates that fewer hours were used. This approach allowed us to evaluate whether our project was completed within the planned human-resource effort.

10.2 Human Resource Effort Estimation

We recorded the planned and actual working hours of each member during the 13-week project period. The recording period is from May 6, 2026, to August 1, 2026, and includes project activities such as project management, requirements analysis, development, testing, documentation, meetings, system integration, and final submission preparation.

Team Member	Main Responsibility	Planned Hours	Actual Hours	Variance
Patricia Orenza	Project management, task coordination, stakeholder communication and submission management	56	45	-11
Lim Yu Jun Felixia	Business analysis, requirement documentation and project-document preparation	54	55	+1
Gu Qiyu	Front-end development for the Mobile App, Admin Web and Landing Web, including front-end integration	56	54	-3
Roy Yoo Hojin	Back-end development, front-end and back-end connection, and system	57	58	+1

	integration			
Vanessa Tan	System testing, defect identification and communication of testing results	50	55	+5
Alvin Rui Long Yeow	System testing, defect identification and communication of testing results	50	53	+3
Total		323	319	-4

We allocated the highest estimated hours to our front-end and back-end development roles because these responsibilities involved the development of multiple technical components and the integration of our system.

We estimated our total human-resource effort at 675 person-hours. These hours represent the value of the time contributed by our members and do not represent salaries paid to them.

10.3 Physical and Technical Resource Estimation

We also used a range of physical and technical resources to complete our project. Although m

Resource	**Quantity or Usage Basis**	**Estimated Project Usage**	**Purpose**
Personal laptops	Six existing laptops	Approximately 600 device-hours	Development, documentation, meetings, testing and final integration
Android testing devices	Existing Android phones used by available team	Two devices	Mobile App installation, interface checking

	members		and demonstration
Android emulator	Development and testing environment	Approximately 40 emulator-hours	Mobile App development and testing when a physical device was unavailable
Combined mobile testing	Android phones and emulator	Approximately 80 device-hours	Installation, functional testing, compatibility checking and demonstration preparation
Internet access	Six members over 10.5 weeks	63 user-weeks	Online meetings, communication, GitHub access, file sharing and software downloads
Microsoft 365, Teams and OneDrive	Six student accounts over 10.5 weeks	63 account-weeks	Communication, meetings, document creation and file storage
GitHub	Shared repository used during development	One shared repository over 10.5 weeks	Source-code storage and version control
Development software	React, React Native, Expo, Node.js, Android Studio, VS Code and related free tools	Used throughout the development period	Front-end, back-end and mobile development
Printing	No project documents were printed	0 pages	Documents were shared and submitted digitally
Project-related transport	Travel for meetings, project work and demonstration activities	Approximately 1–12 one-way journeys	Attendance at necessary face-to-face project activities

We estimated approximately 600 laptop device hours based on the time we spent on development, documentation, testing, and online coordination. Our electricity estimate covered the use of our laptops, Android devices, and emulator-supported activities.

We did not print project materials because we shared and submitted our reports and supporting documents digitally through OneDrive, email, and the submission platform. We only require a limited number of project-related journeys for face-to-face meetings, project coordination, and demonstration activities.

Although our free software and university-supported accounts did not create additional direct cash expenditure, we included their usage because these resources were necessary for us to complete the project.

10.4 Cost Control

We monitored our resource use through weekly meetings, task updates, milestone reviews and checks of our final deliverables. During our meetings, members reported their progress and identified tasks that required additional time.

We used the following measures to control our project resource costs:

1. We used our existing laptops and Android devices instead of purchasing new equipment.
2. We selected free and university-supported software and services wherever possible.
3. We used GitHub to manage our source-code versions and reduce the risk of duplicated or lost development work.
4. We used OneDrive to share documents digitally, removing the need to print project materials.
5. We assigned development and testing responsibilities to specific members to reduce duplicated work.
6. We concentrated additional meetings and integration work around important deadlines and our final demonstration.
7. We required any proposed paid service or additional resource to be discussed with our Project Manager before use.

When a task required more time than we initially expected, our Project Manager could revise the schedule, clarify responsibilities or redistribute work. This was particularly important during our development, testing and final system-integration stages.

11. Project Procurement Management

The concept of project procurement involves the acquisition of any external goods and services required by the project from an outside source. In our project called Secure QR Code Development, all resources that were required were already accessible to us without spending money. We did not purchase any paid software, paid cloud services, or paid hardware. Nevertheless, we recorded those resources which we used and how we accessed them.

Requirements for the Hardware and Software in this Project

Below are listed the hardware and software requirements for the completion of the project. Every team member used his/her own laptop/computer for coding, writing reports, conducting tests, etc. No specific hardware was required to purchase.

The software used by the group includes:

- Microsoft Word – for developing all project documents. It can be obtained from the university student accounts.
- Microsoft Teams and Email – for official correspondence between the team members and client/supervisor. Both programs can be obtained from the university student accounts.
- WhatsApp – for everyday communication between the team members. It is a free application for smartphones.
- OneDrive – for storage and distribution of project documents, minutes of meetings, and reports. They can be accessed through university student accounts.
- GitHub – for storing the code, version control, and task management. Free version is enough for this project.
- Microsoft Project – for creating Gantt chart and project schedule. It can be downloaded from university lab computers/student accounts.
- Microsoft Visio – for developing the network diagram. It can be downloaded from university lab computers/student accounts.
- Node.js, Express, PostgreSQL, React, React Native – for developing the system itself. All the above programs are completely free.
- Postman – for API testing. Free version is enough.

No cloud server hosting, paid databases, domain names, and other similar items will be purchased for this project. Everything will be developed on our computers.

Product Assessment

The Technical Architect and developers, Roy Yoo Hojin and Gu Qiyu, were tasked with assessing the technical products or tools required for development. They first determined whether there was an option available that would allow the product to be acquired free of charge before recommending that the organization pay for anything. In addition, Felixia, the Business Analyst, evaluated the requirements and documentation management tools. The project management tools were assessed by Patricia, the Project Manager.

In view of the fact that this was an academic exercise without funding for any paid services, our team undertook the following process prior to using any paid product:

1. Team members seeking to use a product searched for its free alternative. There are many free products available for development, testing, documentation, and communications purposes.

2. Where there were no free alternatives, the team member submitted a brief justification highlighting reasons for requiring the product; its cost, required period, and inability of free alternatives to meet the purpose.

3. The Project Manager proposed the request in the upcoming team meeting. The project manager, along with some other members of the team, examined the request and determined if it really was necessary to use the product.

4. Where the team decided in favor of using the product, the Project Manager sought the supervisor's permission for purchase.

5. After approval from the supervisor, the team ascertained if the product was really essential for project completion.

Where there were free products, the procurement process was not necessary. It was sufficient for a team member to download and utilize the product; however, they made the other members aware of the product through a WhatsApp message within 24 hours.

When it came to any product that saved data or source code, we ensured it did not contradict our current tools in any way. For instance, where we added a document saving product, it functioned well alongside OneDrive. In the case of a new code saving product, it worked together with GitHub

Final Procurement Summary

Throughout the project, the following tools were successfully used:

Category	Tools Used	Source	Cost
Communication	WhatsApp, Microsoft Teams, Email	Free / University	$0
Documentation	Microsoft Word,	University accounts	$0

	OneDrive		
Project Management	Microsoft Project, Microsoft Excel	University lab computers	$0
Diagramming	Microsoft Visio	University lab computers	$0
Version Control	GitHub	Free account	$0
Development	Node.js, Express, React, React Native, Expo	Open-source / Free	$0
Database	PostgreSQL (Supabase)	Free tier	$0
Testing	Postman	Free version	$0
Code Editing	Visual Studio Code	Free software	$0
Mobile Development	Android Studio, Expo	Free software	$0

The list of tools used during the course of the project was maintained by the Project Manager, who placed it into the OneDrive account and kept it updated when the project adopted new tools. This project did not require any paid tool or service.

12. Conclusion

The Project Management Plan will serve as the primary means of planning and managing the delivery of the Secure QR Code Authentication System to be delivered to the clients, namely, Peter Cole and Mike Groeneweg. The reason why this document was prepared is to integrate all aspects of project management in one coherent plan that all members of the team can adhere to.

The document provided insight into the ten key areas of project management needed in the development of this project. Project Integration Management gave information about teamwork, change management, and the rationale behind two-week sprint cycle. The scope of the project was provided by the Project Scope Management section. The methodology for creating the time management plan was provided by the Project Time Management. The Project Quality Management provided quality assurance measures and indicators used to ensure that all components function as expected. How the team and stakeholder communicated among themselves was detailed in Project Communications Management. Roles and responsibilities of different team members and conflict resolution strategies were indicated in Project Human Resources Management. Project risk management identified 17 potential risks associated with this project. Project cost management assured us that there is no budget attached to this project. Project procurement management detailed the tools used in developing the project and acquiring paid products.

A complete copy of the Work Breakdown Structure, the Gantt chart, the network diagram, Team Charter, glossary of terms, and agenda and minutes of meetings with the client and supervisor are available in the appendices attached at the back of this document.

This document will keep on changing as time progresses, and team members will review it periodically during the entire project. Any changes that occur in the plan will have to go through the change management process discussed in section four. The deliverables for the entire project are expected to be delivered as per the timeline described in the Project Management Plan, with the last deliverable due by 1st August 2026.

13. Appendices

13.1 Appendix A: Deliverable Task Breakdown Statement

Deliverable	Key Tasks	Responsible Member	Acceptance Criteria
Team Charter	Define roles, communication rules, project objectives	All Team Members	Charter approved and submitted
Project Management Plan	Scope, schedule, risk, quality, communications planning	Patricia	PMP submitted by due date
Requirements & Analysis Document	Gather and document functional and non-functional requirements	Felixia	Requirements approved by client and supervisor
Design Document	Architecture design, database design, API design	Roy, Qiyu	Design document completed and submitted
Backend Development	QR generation, verification, alerts APIs	Roy, Qiyu	Backend functions correctly and passes testing
Web Development	Landing page, QR display page, admin dashboard	Roy, Qiyu	Pages function according to requirements
Mobile Application	QR scanner, verification, tamper reporting	Roy, Qiyu	Application functions correctly on Android devices
Testing	Unit, integration, system and user acceptance testing	Vanessa	Test report completed
Documentation	User manual, deployment guide, final report	Alvin, Felixia	Documentation completed and submitted

Final Submission	Compile all deliverables and submit	All Team Members	All project deliverables submitted by deadline

13.2 Appendix B: Work Breakdown Structure and Schedule

WBS (Work Breakdown Structure)

1. Project Management

1.1 Develop and maintain Project Management Plan

1.2 Conduct and document client meetings

1.3 Manage team task allocation and progress tracking

1.4 Conduct risk identification and mitigation planning

1.5 Prepare and submit milestone deliverables

1.6 Maintain the Design Change Register and obtain client approval for behaviour-changing modifications (notably the app-required access model)

2. Requirements and Analysis

2.1 Gather and document functional requirements from client brief

2.2 Identify and document non-functional requirements

2.3 Clarify open items with client (Peter and Mike)

2.4 Produce and submit Requirements and Analysis Document

2.5 Obtain client sign-off on requirements

2.6 Amend the actor definitions for the introduction of user accounts, and re-scope the destination allowlist and GPS manual-entry fallback requirements

3. System Architecture and Design

3.1 Define overall system architecture (backend, web surfaces, mobile app)

3.2 Design database schema (users, QR codes, scan logs, alerts, activity log)

3.3 Define API endpoint structure and contracts

3.4 Define token signing and validation approach (separate QR and authentication JWT secrets, plus session revocation)

3.5 Define the channel identification approach for app and browser users, and the authenticated endpoint used by the app

3.6 Define the two web surface flows and their purposes

3.7 Produce Architecture Design Document

4. Backend Development

- 4.1 Set up Node.js + Express project and managed PostgreSQL database
- 4.2 Implement database schema using Prisma ORM
- 4.3 Implement QR code generation endpoint (POST /api/qr/generate), restricted to administrators, accepting destination URL, expiry, and optional label
- 4.4 Implement token signing service using JWT, with separate QR and authentication secrets
- 4.5 Implement QR verification endpoints
  - 4.5.1 Token validation (signature, expiry)
  - 4.5.2 Blacklist and status check
  - 4.5.3 Browser response branch — verification result and app download prompt, with the destination URL withheld
  - 4.5.4 Scan logging to database on every attempt, including failures and forged tokens
  - 4.5.5 Authenticated mobile verification endpoint attributing each scan to the signed-in account
  - 4.5.6 Lazy expiry sweep flipping time-expired codes to the expired state
- 4.6 Implement alert submission endpoint (POST /api/alert/report), requiring authentication
  - 4.6.1 Photo upload handling to cloud object storage with size and MIME validation, storing only the URL
  - 4.6.2 GPS and description storage
  - 4.6.3 Automatic QR code status update to suspicious, permitted only from the active state
- 4.7 Implement authentication endpoints and access-control middleware
  - 4.7.1 Registration and login with bcrypt password hashing
  - 4.7.2 Forgot-password and reset-password with single-use expiring tokens
  - 4.7.3 Logged-in password change and session revocation
  - 4.7.4 Authenticated-user and administrator-only middleware
  - 4.7.5 First-administrator seed script driven by environment variables
- 4.8 Implement admin API endpoints
  - 4.8.1 Get all QR codes with search, status filter, pagination, and summary counts
  - 4.8.2 Get individual QR code details with scan history and an on-demand regenerated QR image
  - 4.8.3 Update QR code status (blacklist / activate), rejecting changes to time-expired codes
  - 4.8.4 Get all tamper alerts and resolve them
  - 4.8.5 Get metrics and telemetry summary

4.8.6 User administration — list, create, change role with the last-administrator guard, and delete

4.8.7 Recent activity feed and CSV export of the QR code register

4.9 Implement per-user scan history endpoint for the mobile application

5. Web Page Development

5.1 QR Code Landing Page (for regular scanner users)

5.1.1 Display token validation result (safe / warning / blacklisted / expired)

5.1.2 Display the app download prompt and APK link — no destination URL and no continue option is offered to browser users

5.1.3 Display a pop-up explaining that the dedicated app is required if the user declines the download

5.1.4 Display clear warning message for invalid or blacklisted codes

5.1.5 Provide a mobile-responsive layout

5.2 QR Code Display function (integrated into the admin dashboard)

5.2.1 Display the generated QR code image in the generation-result modal

5.2.2 Show destination URL, label, creation and expiry information, status, and total scans in the QR code detail view

5.2.3 Provide download and print options, with the image regenerated on demand from the stored token

5.3 Admin Management Dashboard

5.3.1 Implement application layout and navigation sidebar

5.3.2 Implement Dashboard page with metrics cards, scan volume chart, status distribution chart, and recent activity feed

5.3.3 Implement QR Code Management page with search, filtering, pagination, generation form, and CSV export

5.3.4 Implement Alerts review page with photo, GPS, description, reporter details, and resolve action

5.3.5 Implement Users page with search, role change, account creation, and deletion

5.3.6 Implement Settings page with profile editing and password change

5.3.7 Implement authentication pages and route guards for all dashboard routes

5.3.8 Connect all pages to backend AP

6. Mobile Application Development

6.1 Set up React Native project

6.2 Implement login and registration screens, with registration confirming account creation and requiring an explicit sign-in

6.3 Implement QR code scanner screen

6.4 Implement token submission to the authenticated backend verification endpoint

6.5 Implement verification result screen (safe / warning) and destination access for valid codes

6.6 Implement tamper alert reporting screen

- 6.6.1 GPS location capture
- 6.6.2 In-app camera for photo capture
- 6.6.3 Description and optional contact form
- 6.6.4 Alert submission to backend

6.7 Implement personal scan history screen backed by the server

6.8 Set custom User-Agent header (SecureQRApp/1.0) on all requests for channel telemetry

6.9 Build and publish the Android APK and configure the download link used by the landing page

7. Testing

7.1 Develop test plan covering all functional requirements, re-baselined for the app-required access model

7.2 Unit testing of backend services (token, QR generation, alert)

7.3 Integration testing of all API endpoints, including authentication and role-based access control

7.4 End-to-end testing of the full scan and alert flow

7.5 Backward compatibility testing with standard QR scanners

7.6 Web page testing across mobile and desktop browsers

7.7 User acceptance testing with client

7.8 Document and report test results

8. Documentation

8.1 Write and maintain Requirements and Analysis Document

8.2 Write Architecture Design Document

8.3 Write User Manual for admin dashboard and mobile app, including the user administration chapter

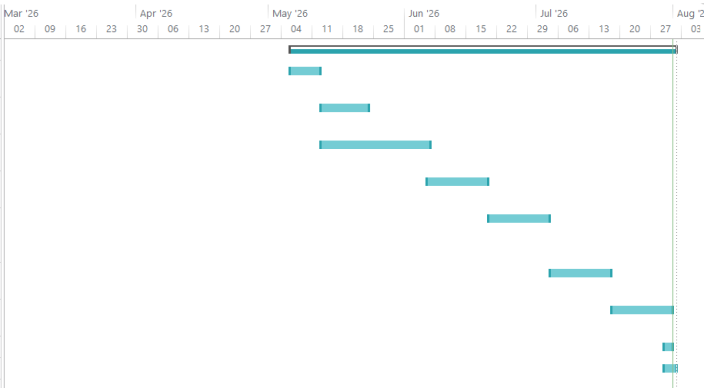
8.4 Write Installation and Deployment Guide covering database connection rules, build and deployment commands, and file storage configuration

8.5 Maintain the Design Change Register and record the known limitations of the delivered system

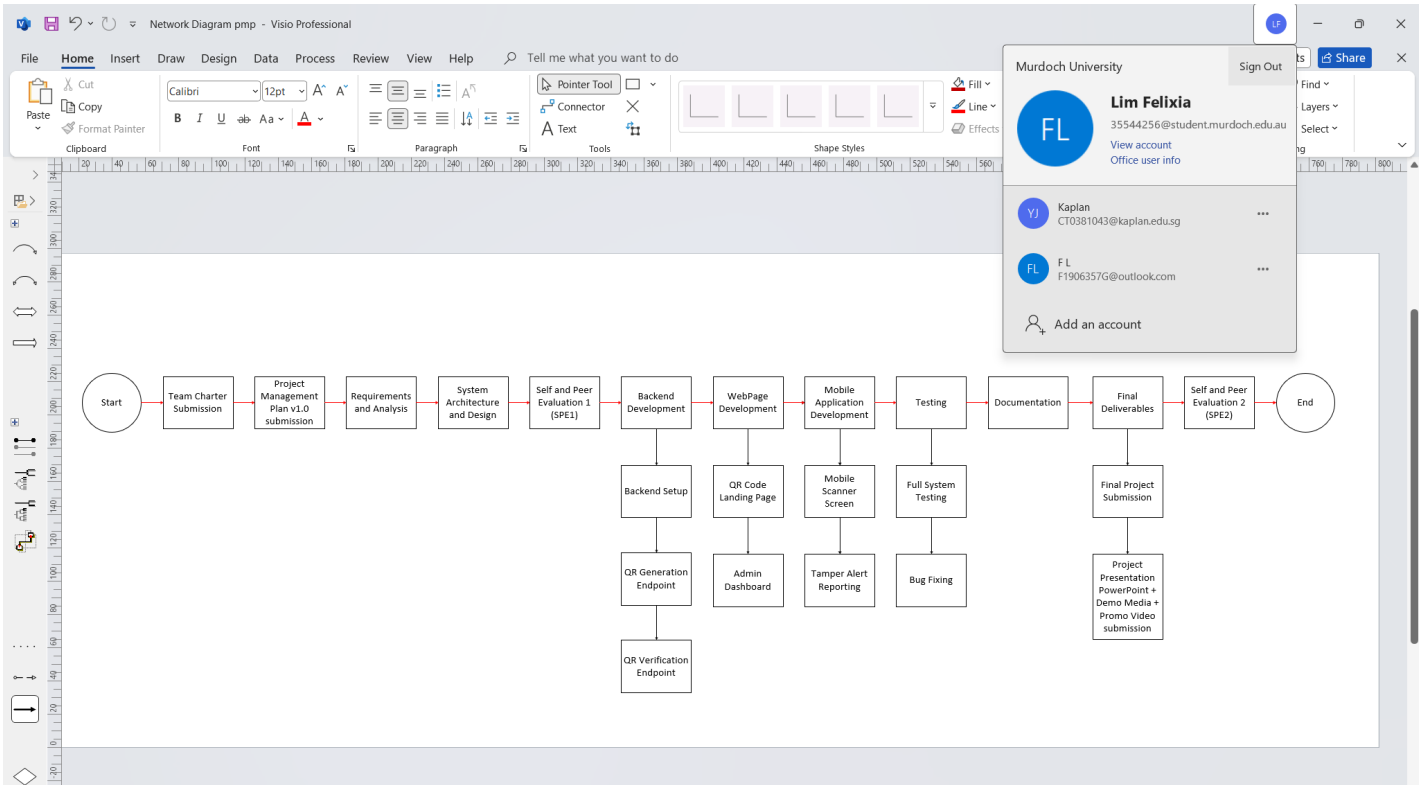
8.6 Produce Final Project Report

5.3 Gantt Chart

Task Name	Start	Finish	Duration	Predecessors
VAFPQR	**Wed 06/05/26**	**Sat 01/08/26**	**64 days**	
Sprint 1: Group Allocation	Wed 06/05/26	Tue 12/05/26	5 days	
Sprint 2: Roles & Team Charter	Wed 13/05/26	Sat 23/05/26	9 days	
Sprint 3: PMP & Requirements	Wed 13/05/26	Sat 06/06/26	19 days	
Sprint 4: R&A, Design, Setup	Sat 06/06/26	Fri 19/06/26	11 days	
Sprint 5: Design Completion, QR Endpoints	Sat 20/06/26	Fri 03/07/26	11 days	
Sprint 6: Pages, APK, Scanner, Alerts	Sat 04/07/26	Fri 17/07/26	11 days	
Sprint 7: Demo, Testing, Docs,	Sat 18/07/26	Fri 31/07/26	11 days	
Buffer Days	Thu 30/07/26	Fri 31/07/26	2 days	
Final Submission	Thu 30/07/26	Sat 01/08/26	3 days	



5.4 Network Diagram



13.3 Appendix C – Project/Team charter

Project Objectives:

To create and build a secure authentication and management system for QR codes that:

- Stops and identifies tampering with publicly accessible QR codes.
- Delivers a management system featuring expiration, distinct authorization codes, redirects, and tracking.
- Features a mobile application for scanning and verifying QR codes. Stays compatible with regular QR scanners (iOS/Android native).
- Allows users to report disruptions related to location, images, and optional contact details.

Approach: Summary of the stakeholder's needs and expectations, important assumptions, constraints. Refer to Project Management Plan and Communication Management Plan.

- We will meet every week to check our progress and plan the next steps.
- Each meeting will last around 2 to 3 hours.
- Between meetings, we will work on our own tasks and message each other on WhatsApp if we have questions.

Stakeholders: Peter Cole and Mike Groeneweg (client)

Assumptions:

- Users have smartphones with cameras and GPS.
- QR codes only for verified users and only able to access through the mobile application me made.
-

Constraints:

- Time: Complete by the end of TMA2026
- Must work on android
- Skills: not everyone knows how to do mobile development need to learn new skills and research

Roles and Responsibilities:

Role	Name	Position/ Key Responsibilities
Project Manager	Patricia Mikaela Marie Jusay Orenza	• Create and maintain the project schedule with clear deadlines. • Break down, assign work so that everyone has equal work • Track progress of all team members and remind them of upcoming deadlines. • Be the main point of contact for the clients Peter Cole and Mike Groeneweg. • Keep a list of potential problems (risk register) and update it. • Spot issues early before they become big problems. • Escalate issues to lecturer and in charge immediately • Make sure the team meets all project requirements. • Check that deliverables are submitted on time and in the correct format. • Help resolve conflicts between team members. • Make sure everyone has what they need to do their work. • Step in to help any role if someone is falling behind.
Business Analyst	Lim Yu Jun Felixia	• Understanding requirements and record down • Turn client needs into simple "user stories" • Write clear rules • Manage any changes or new requests

		from clients • Help translate between business needs and technical tasks • Work with the tester to make sure everything is tested • Write user manuals from a beginner's perspective
Technical Architect/ Developer	Roy Yoo Hojin, Gu QiYu	• Build the admin dashboard to see all QR codes (Show statistics: number of scans, last scan time, locations on a map) • Build the redirect system: direct user to make sure they scan the qr code using the mobile application • Store all scan data into database with consolidated/comprehensive report
Documentation	Alvin Rui Long Yeow,	• Develop QR code generator system • Document minutes for meeting • Collaborate with tester on her key responsibilities
Tester	Vanessa Tan	• Develop QR code generator system including specified features (expiry, redirect, geoloc etc) • Testing to ensure that generated QR codes and mobile application is functional. • Verifying the QR code authentication, redirection and expiry timeout. • Identifying bugs and features and recommending security measures and improvements. • Help to resolve issues and ensure system reliability, security and performance.

Our group's name: VAFPQR

Our goals

What is our group trying to accomplish?

1. Build a working system by the end of the teaching period

2. Make sure QR code and management platform is accessible and working
3. Able to work with Android

Our ground rules

- When will we attempt to meet (what time, how often, how long)?

  Meet twice every week

- Where will our meetings take place?

  - Thursday 7-8pm / 8-9pm

  - Friday 7-8pm / 8-9pm

- How do we inform each other when we cannot be there or are running late? How else can we communicate and keep others up to date?

  1. Notes and minutes taken every meeting
  2. Inform at least 3 hours earlier if unable to attend meeting with reasons.
  3. Take initiative to follow up with groupmates if you miss out any information.
  4. If any conflict votes for an agreement.

- How do we deal with lateness to meetings? What does “on time” mean?

  On time meeting time as discussed. 10min delay accepted unless informed in advance otherwise considered late.

- How do we deal with members who don’t participate enough, participate too much, or distract the group from its task?

  2 warnings if still no change inform lecturer.

- How are we going to make decisions?

  1. Discuss with team make sure everyone in on par and agrees to it
  2. Any **conflicts**, vote and follow majority
  3. Try to settle as soon as possible.

- What will we do if a group member’s work doesn’t meet our standards?

  1. Give feedback and suggestion on how we can improve together.
  2. Give support and help.
  3. Open to communication.

13.4 Appendix D – Glossary of terms

Term	Definition
QR Code	Quick Response Code
JWT	JSON Web Token
API	Application Programming Interface
UAT	User Acceptance Testing
GPS	Global Positioning System

13.5 Appendix E – Agenda and minutes of all client meetings

Chairperson: Patricia

Secretary: Felixia

Agenda

Item No.	Description of Item	Introduced by	Approx. time for item
1.	User validation and authentication requirements	Team	10 minutes
2.	Security and encryption necessities	Team	5 minutes
3.	User and system process workflow	Team	10 minutes
4.	QR code scanning and validation	Team	15 minutes
5.	Future steps and confirmation of project requirements	Team	5 minutes
Total time		45 minutes	

Meeting minutes

Meeting date: 26/5/26

Meeting commenced at: 1.05 pm

Meeting concluded at: 1.55 pm

Members present:

1. Vanessa Tan [redacted]
2. Patricia Orenza [redacted]
3. Lim Yu Jun Felixia [redacted]
4. Roy Hojin Yoo [redacted]

1. Authentication and validation

Authentication based on User ID was said to be suitable for the current phase of the project.

Action to be taken	Person/s responsible	Action to be completed by
Explore more on the user ID authentication approach.	Roy Yoo Hojin and Gu Qiyu	3/6/26

2. Authentication validation

Client stated that the solution does not necessarily have to be a dedicated QR scanning application, browser, OS-based and other existing scanners will be further explored.

Action to be taken	Person/s responsible	Action to be completed by
QR validation methods investigation	Roy Yoo Hojin and Gu Qiyu	3/6/26

3. Encryption and security

Encryption of data relating to QR codes and validation processes were addressed but the definitive implementation strategy is yet to be decided.

Action to be taken	Person/s responsible	Action to be completed by
Research on different types of encryptions methods for QR data and determine what is most ideal.	Vanessa Tan, Alvin Rui Long Yeow	3/6/26

4. User workflow and process workflow

Our proposed workflow was reviewed and accepted as a baseline structure.

Action to be taken	Person/s responsible	Action to be completed by
Enhance and document final system process	Lim Yu Jun Felixia	20/5/26

5. Documentation

The outcomes of meeting and project decisions will be recorded and stored for tracking of project.

Action to be taken	Person/s responsible	Action to be completed by
Keep track and document meeting minutes	Alvin Rui Long Yeow	20/5/26

6. Testing preparation

Requirements regarding testing were discussed briefly to make sure that the system's functionality aligns with the expectations before committing.

Action to be taken	Person/s responsible	Action to be completed by
Understand and prepare potential test cases and review the system's functionality.	Vanessa Tan	20/5/26

13.6 Appendix F – Agenda and minutes of all supervisor meetings

Chairperson: Patricia

Secretary: Felixia

Week 2

Agenda

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Get to know more about team members	Team	10 minutes
2.	Understand the roles available	Team	10 minutes
Total time		20 minutes	

Meeting minutes

Meeting date: 13/5/26

Meeting commenced at: 10.15 am

Meeting concluded at: 10.40 am

Members present:

1. Vanessa Tan [redacted]
2. Alvin Yeow [redacted]
3. Patricia Orenza [redacted]
4. Lim Yu Jun Felixia [redacted]
5. Gu Qiyu [redacted]
6. Roy Hojin Yoo [redacted]

1. Organisation of team and mapping skills

We were just getting to know more about each other, understanding each other's skills and majors.

Action to be taken	Person/s responsible	Action to be completed by
Examining individual strengths to ensure balanced work distribution (Writing, documentation, technical)	All members	13/5/26

Allocating team roles (Project manager, business analyst, programmer, tester)	All members	13/5/26

2. Operational criteria and communication

We discussed and planned on ways to communicate with each other and when does everyone’s schedule align for further meetings.

Action to be taken	**Person/s responsible**	**Action to be completed by**
Establishing a collaborative digital workspace (eg. Teams, OneDrive, Google drive)	Patricia (Project Manager)	**13/5/26**
Discussion and planning of weekly meeting time	All members	**13/5/26**

Chairperson: Patricia

Secretary: Felixia

Week 3

Agenda

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Discussion for preparation of meeting with client	Team	10 minutes
2.	More specific delegation of duties to members	Team	20 minutes
3.	Completion of Team Charter	Team	20 Minutes
Total Time		50 Minutes	

Meeting minutes

Meeting date: 20/5/26

Meeting commenced at: 10.25 am

Meeting concluded at: 10.50 am

Members present:

1. Vanessa Tan [redacted]
2. Alvin Yeow [redacted]
3. Patricia Orenza [redacted]
4. Lim Yu Jun Felixia [redacted]
5. Roy Hojin Yoo [redacted]

1. Client Preparation and Internal task delegation

Discussed what each member should do to prepare for the meeting with client

Action to be taken	Person/s responsible	Action to be completed by
Team discussion to prepare for meeting with client	All members	20/5/26

More specific delegation of tasks to each member to ensure a smoother workflow

Action to be taken	Person/s responsible	Action to be completed by
More specific delegation of duties to members	All members	20/5/26

2. Team completion of Team Charter

Discussion among team members to complete Team Charter

Action to be taken	Person/s responsible	Action to be completed by
Completion of Team Charter	All members	24/5/26

Week 4

Agenda

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Discuss and share what we found out after research on existing QR systems	Team	20 minutes
2.	Additional preparation for client meeting	Team	15 minutes
3.	Brainstorming session for non-functional components of the project	Team	10 minutes
Total time		45 minutes	

Meeting minutes

Meeting date: 26/5/26

Meeting commenced at: 10.30 am

Meeting concluded at: 11 am

Members present:

1. Vanessa Tan [redacted]
2. Lim Yu Jun Felixia [redacted]
3. Roy Hojin Yoo [redacted]

1. Updates before client meeting

After further research and discussions, we came up with more resources and questions to affirm our curiosity about the client's expectations.

Action to be taken	Person/s responsible	Action to be completed by
Update questions to confirm with the client.	All members	26/5/26
Create a draft process flow chart to let the client know our outline of the project.	All members	26/5/26

Week 5

Agenda

Item No.	Description of Item	Introduced by	Approx. time for item
1.	Confirming what software and tools we will be using for the QR code scanning and validation.	Roy, Qiyu	15 minutes
2.	Figuring out what security measures can be implemented with the software used	Vanessa, Alvin	15 minutes
3.	Come up with a more detailed structure of process to maximise time	Felixia, Patricia	15 minutes
4.	Update each other on individual process	Team	10 minutes
5.	Developing testing approach for system verification	Vanessa, Alvin	10 minutes
Total time		65 minutes	

Meeting minutes

Meeting date: 3/6/26

Meeting commenced at: 10.45 am

Meeting concluded at: 11.25 am

Members present:

1. Vanessa Tan [redacted]
2. Lim Yu Jun Felixia [redacted]
3. Gu Qiyu [redacted]
4. Roy Hojin Yoo [redacted]

1. Tools and software for QR system

Confirmation of software and tools that will be used for this project is discussed and further explanation of why and how they work.

Action to be taken	Person/s responsible	Action to be completed by
Proceed with development utilizing the tools and software	Roy, Qiyu	Ongoing

2. Security measures

Security protocols such as encryption methods relating to QR were reviewed

Action to be taken	Person/s responsible	Action to be completed by
Implementation plan for security and encryption methods	Vanessa, Alvin	Ongoing

3. System and workflow process structure

A more detailed process structure was discussed to ensure that workflow is efficient and equal.

Action to be taken	Person/s responsible	Action to be completed by
Follow and implement the updated system workflow.	Felixia, Patricia	Ongoing

4. Progress updates

Each team members to update their personal progress accordingly to roles assigned.

Action to be taken	Person/s responsible	Action to be completed by
Continue in developing accordingly to the roles assigned	All members	Ongoing